SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1701**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*
Assessment Tool Weightage: 50%

QUESTIONS: 3 **Duration : 1 Hour**
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:
I JAYA PRAKASH M(1912260496) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student:JAYAPRAKASH

Annexure A

Constraint-Based Problem Solving

1.A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

Constraints:

i. D1 cannot work Night shift
ii. D2 must work before D3 (Morning < Afternoon < Night)
iii. Only one doctor per shift
iv. D3 cannot work Morning
v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2.A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

i. Movement allowed: Up, Down, Left, Right
ii. Cost of each move = 1
iii. Cannot pass through obstacles
iv. Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost**.**

3.An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.
The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:
a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality
- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

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ASSESSMENT TOOL 2

COURSE: ARTIFICIAL INTELLIGENCE (CSA1701)

Name	JAYA PRAKASH M
Register No.	1912260496

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.

CONSTRAINT-BASED PROBLEM SOLVING

Question 1 – Backtracking Search

Given: Doctors D1, D2, D3 and shifts Morning (M), Afternoon (A), Night (N).

- D1 cannot work Night.
- D2 must work before D3: Morning < Afternoon < Night.
- Only one doctor can work in each shift.
- D3 cannot work Morning.
- Every doctor must be assigned exactly one shift.

Backtracking steps:

1. Start with an empty assignment: { }.
2. Try D1 = Morning. This satisfies the D1 restriction.
3. Try D2 = Afternoon. Since D2 is before D3, D3 must be after Afternoon, so D3 = Night.
4. Check all constraints: D1 = Morning, D2 = Afternoon, D3 = Night. D1 is not Night; D2 is before D3; shifts are unique; D3 is not Morning; and all doctors are assigned.
5. Therefore, no backtracking is required after this successful branch.

Constraint-check table:

Assignment	Constraint check	Result
D1 = Morning	D1 is not Night	✓
D2 = Afternoon	D2 is before D3	✓
D3 = Night	D3 is not Morning	✓
All shifts	One doctor per shift	✓
All doctors	Exactly one shift each	✓

Final valid schedule: Morning – D1; Afternoon – D2; Night – D3.

Question 2 – Grid Search

The assessment specifies a 5×5 grid, four-direction movement, unit movement cost, obstacles, and Manhattan distance heuristic. However, the uploaded question paper does not provide the actual positions of S, G, or the obstacles. Therefore, a unique numerical optimal path and cost cannot be calculated from the supplied paper without inventing missing data.

Correct search procedure:

1. Place the Start node S in the search structure with path cost g(S) = 0.
2. Generate only valid neighboring cells: Up, Down, Left and Right.
3. Reject any neighbor that is outside the 5×5 grid or is an obstacle.
4. For BFS, expand nodes level-by-level using a FIFO queue. Because every move costs 1, the first time the goal is reached, the path has minimum number of moves.
5. If the Manhattan heuristic $h(n) = |x_n - x_G| + |y_n - y_G|$ is used in a priority queue, the procedure is A* search rather than ordinary BFS. Use $f(n) = g(n) + h(n)$ for priority.
6. Continue expansion until G is selected. Trace parent links from G back to S to obtain the optimal path.

Manhattan heuristic: $h(n) = |x - xG| + |y - yG|$.

Cost: Since each movement costs 1, total path cost equals the number of moves in the final path.

Question 3 – Autonomous Rescue Robot

3(a) Analysis of constraints

- Four-direction movement: The robot can move only Up, Down, Left and Right, so diagonal movement is not allowed.
- Obstacles: Blocked cells cannot be entered and must be excluded from the search.
- Partial observability: The robot can observe only adjacent cells, so it does not initially know the complete environment.
- Movement cost: Each normal movement costs 1.
- Risky zones: Entering a risky zone adds 2 extra cost, making its movement cost 3.
- Online knowledge update: The robot must update its internal map whenever it senses new information.
- Objective: Reach G while minimizing total cost and maintaining safe navigation.

3(b) Solution approach – Uniform Cost Search with Online Updates

1. Initialize the robot at S with path cost 0 and store the currently known map.
2. Sense the adjacent cells. Mark newly observed obstacles and risky cells in the robot's knowledge base.
3. Generate safe, traversable neighboring cells.
4. Assign cost 1 to a normal move and cost 3 to a move into a risky zone (1 + additional risk cost 2).
5. Insert available nodes into a priority queue ordered by accumulated path cost g(n).
6. Expand the node with the lowest accumulated cost. This is the principle of Uniform Cost Search.
7. Whenever new information is sensed, update the map and revise the available search choices if necessary.
8. Continue until the robot reaches G. The selected path has minimum known total cost among the feasible paths.
9. If the environment changes or an unexpected obstacle is discovered, re-plan from the robot's current position using the updated knowledge.

3(c) Application of AI concepts

Intelligent Agent: The rescue robot acts as an intelligent agent because it perceives its surroundings through sensors, maintains knowledge about the environment, selects actions, and acts to reach the survivor.

Rationality: A rational robot chooses actions that best achieve its goal using the information currently available. It avoids blocked cells and prefers a lower-cost safe route.

Environment Type: The environment is partially observable, sequential, dynamic/uncertain from the robot's limited viewpoint, and has different action costs because risky zones add cost.

3(d) Justification

Uniform Cost Search is suitable because the problem has non-uniform path costs: a normal move costs 1, while entering a risky zone costs 3. UCS expands the least-cost path first and therefore is appropriate for minimizing total cost. Since the robot cannot see the complete environment in advance, the online-search behavior allows it to update its knowledge and re-plan when new obstacles or risks are discovered. This combination supports safe and cost-effective navigation under partial observability.

Conclusion

The problems demonstrate important AI concepts including constraint satisfaction, backtracking, search strategies, intelligent agents, rational decision-making, and online search. The solutions satisfy the stated constraints while avoiding assumptions where the assessment paper does not provide required grid data.

Student Signature: JAYAPRAKASH